

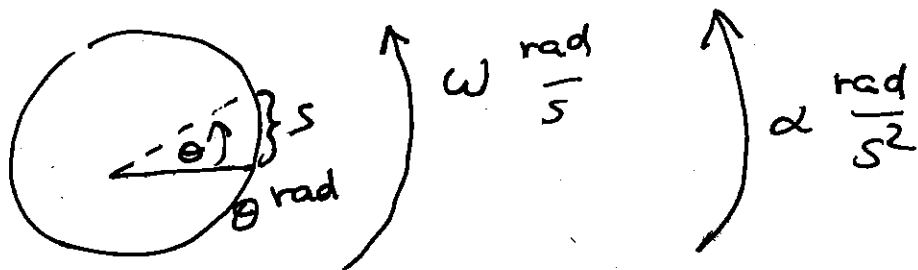
10 Rotation

1/8

$$s = R\theta$$

$$v = R\omega$$

$$a = R\alpha$$



$$F = ma \rightarrow \tau = I\alpha \quad \vec{\tau} = \vec{r} \times \vec{F}$$



$$K_{\text{trans}} = \frac{1}{2}mv^2 \rightarrow K_{\text{rot}} = \frac{1}{2}I\omega^2$$

$$p = mv \rightarrow L = I\omega$$

$$F = \frac{dp}{dt} \rightarrow \tau = \frac{dL}{dt}$$

$$v_f = v_i + at$$

$$\rightarrow \omega_f = \omega_i + \alpha t$$

$$x_f = x_i + \frac{v_i + v_f}{2} t$$

$$\rightarrow \theta_f = \theta_i + \frac{\omega_i + \omega_f}{2} t$$

$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$\rightarrow \omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i)$$

$$x_f = x_i + v_i t + \frac{1}{2}at^2$$

$$\rightarrow \theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2$$

Moment of Inertia

2/8

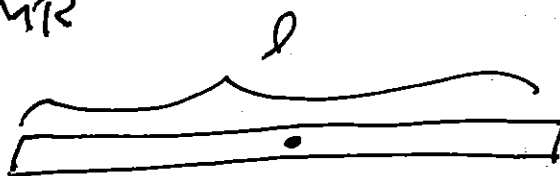
I is resistance to rotation.

Ring $I_{\text{ring}} = MR^2$

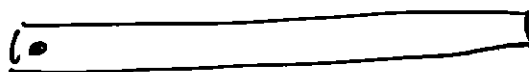
DISK $I_{\text{disk}} = \frac{1}{2} MR^2$

SPHERE $I_{\text{sphere}} = \frac{2}{5} MR^2$

$I_{\text{rod}} = \frac{1}{12} ml^2$
middle



$I_{\text{rod}} = \frac{1}{3} ml^2$
end



parallel axis theorem $I = I_0 + md^2$

I_0 moment about its center axis

$d = l/2$ d distance axis is moved



$$I = I_0 + md^2$$

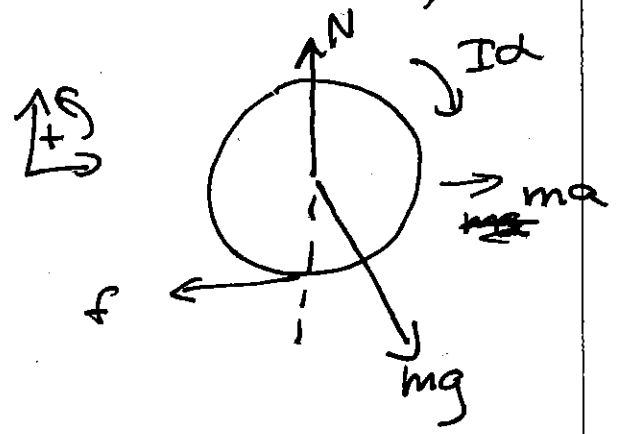
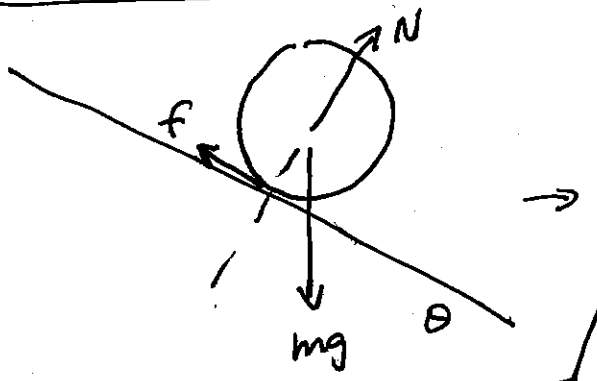
$$= \frac{1}{12} ml^2 + m\left(\frac{l}{2}\right)^2$$

$$= \left(\frac{1}{12} + \frac{1}{4}\right) ml^2 = \left(\frac{1}{12} + \frac{3}{12}\right) ml^2$$

$$= \frac{1}{3} ml^2$$

$$I = \sum m_i r_i^2 = \int r^2 dm$$

3/8



$$\sum F_x: mg \sin \theta - f = ma$$

$$\sum \tau_z: -Rf = -I\alpha =$$

$$Rf = \frac{2}{5} MR^2 \alpha$$

$$f = \frac{2}{5} MR \alpha$$

$$R\alpha = a$$

$$f = \frac{2}{5} Ma$$

$$mg \sin \theta - \frac{2}{5} ma = ma$$

$$g \sin \theta = \left(1 + \frac{2}{5}\right) a = \frac{7}{5} a$$

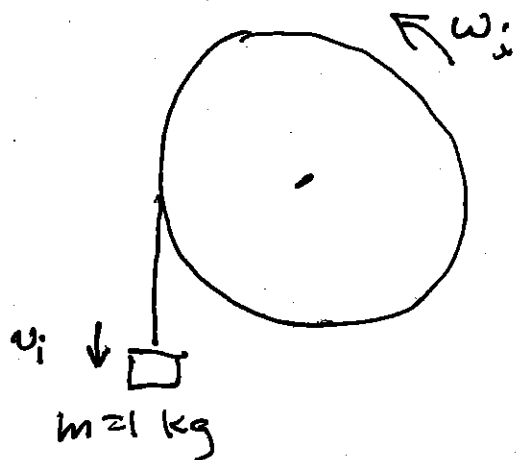
$$a = \frac{5}{7} g \sin \theta$$

less than sliding

$$a_{\text{slide}} = g \sin \theta$$

Atwood's Machine w/ torque

4/9



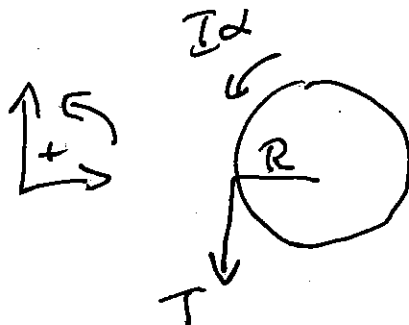
pulley I, M, R

$$\text{disk } I = \frac{1}{2} MR^2 \quad M = 10 \text{ kg}$$

$$R = 0.5 \text{ m}$$

$$v_i = \text{---} 1 \text{ m/s down}$$

$$v_f = ? \text{ when } \Delta x = 2 \text{ m}$$



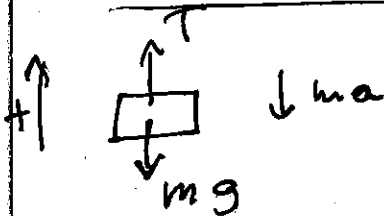
$$RT = I\alpha$$

$$RT = \frac{1}{2} MR^2 \alpha$$

$$T = \frac{1}{2} MR \alpha$$

$$R\alpha = a$$

$$\boxed{T = \frac{1}{2} Ma}$$



$$T - mg = -ma$$

$$\frac{1}{2} Ma - mg = -ma$$

$$mg = \left(m + \frac{M}{2}\right) a$$

$$a = \frac{mg}{m + \frac{M}{2}} = \frac{9.8}{1 + \frac{10}{2}} = 1.63 \frac{\text{m}}{\text{s}^2} \text{ down}$$

$$v_f^2 = v_i^2 + 2a(y_f - y_i)$$

5/8

$$v_f^2 = (-1)^2 + 2(-1.63)(0-2)$$

$$= 1 + 2(-1.63)(-2)$$

$$= 1 + 6.52 = 7.52$$

$$v_f = \sqrt{7.52} = 2.74 \text{ m/s down}$$

Choose - root

NOW BY CONS OF ENERGY

$$E_i = E_f$$

$$\frac{1}{2}mv_i^2 + \frac{1}{2}I\omega_i^2 + mgh = \frac{1}{2}mv_f^2 + \frac{1}{2}I\omega_f^2$$

$$\frac{1}{2}mv_i^2 + \frac{1}{2}\frac{1}{2}MR^2\omega_i^2 + mgh = \frac{1}{2}mv_f^2 + \frac{1}{2}\frac{1}{2}MR^2\omega_f^2$$

$$\left(\frac{1}{2}m + \frac{1}{4}M\right)v_i^2 + mgh = \left(\frac{1}{2}m + \frac{1}{4}M\right)v_f^2$$

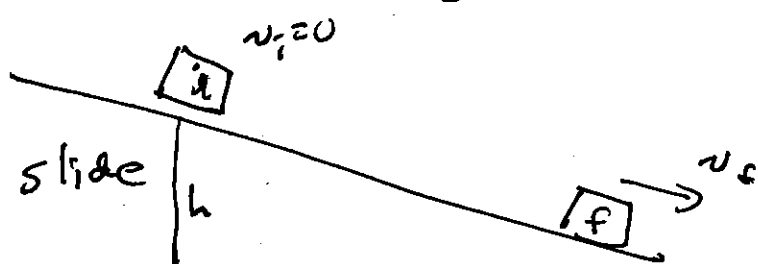
$$v_f = \sqrt{\frac{\left(\frac{m}{2} + \frac{M}{4}\right)v_i^2 + mgh}{\frac{m}{2} + \frac{M}{4}}}$$

$$v_f = \sqrt{\frac{\left(\frac{1}{2} + \frac{10}{4}\right)(-1)^2 + 1 \cdot 9.8 \cdot 2}{\frac{1}{2} + \frac{10}{4}}}$$

$$v_f = \sqrt{\frac{3 + 19.6}{3}} = \boxed{2.74 \text{ m/s}}$$

6/8

Consider a mass rolling down an incline from a height h .

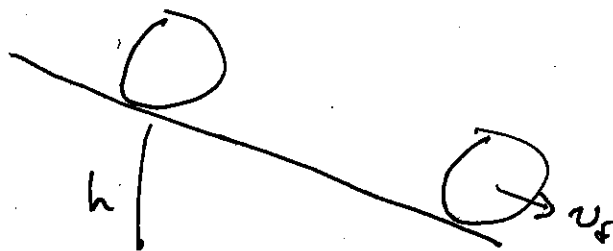


$$mgh = \frac{1}{2} mv^2$$

$$v = \sqrt{2gh}$$

Ball rolls down incline

$$I = \frac{2}{5} mr^2$$



$$mgh = \frac{1}{2} mv^2 + \frac{1}{2} I\omega^2$$

$$mgh = \frac{1}{2} mv^2 + \frac{1}{2} \frac{2}{5} mr^2 \omega^2$$

$$r\omega = v$$

$$mgh = \frac{1}{2} mv^2 + \frac{1}{5} mv^2 = \left(\frac{1}{2} + \frac{1}{5}\right) mv^2$$

$$gh = \frac{7}{10} v^2$$

$$v = \sqrt{\frac{10}{7} gh} = \sqrt{1.43 gh}$$

Disk rolls down

7/8

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$I = \frac{1}{2}mr^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{4}mr^2\omega^2$$

$$r\omega = v$$

$$mgh = \cancel{\frac{1}{2}mv^2} + \frac{1}{2}mv^2 + \frac{1}{4}mv^2$$

$$mgh = \frac{3}{4}mv^2$$

$$gh = \frac{3}{4}v^2$$

$$v = \sqrt{\frac{4}{3}gh} = 1.33\sqrt{gh}$$

Ring rolls down

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$I = mr^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}mv^2$$

$$mgh = mv^2$$

$$gh = v^2$$

$$v = \sqrt{gh}$$

8/8

obj	symbols	numbers
ring	$v = \sqrt{gh}$	$v = \sqrt{1.0 gh}$
disk	$v = \sqrt{\frac{4}{3} gh}$	$v = \sqrt{1.33 gh}$
sphere	$v = \sqrt{\frac{10}{7} gh}$	$v = \sqrt{1.43 gh}$
slide	$v = \sqrt{2 gh}$	$v = \sqrt{2 gh}$